Too many cooks spoil the broth?

Exploring the challenges for collaboration between museums and ICT design companies

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Abstract: To collaborate across organisational contexts is not an easy task, but to do so and to succeed can be vital for reaching creative and innovative results. This paper is an empirical exploration of contextual challenges for collaborative museum design, focusing on the interplay between museums and ICT design companies. Based on empirical findings, two kinds of challenges are presented as particularly interesting in relation to this context: Structure related challenges, and value and practice related challenges.

Keywords: collaborative design, ICT design, exhibition design, design processes, ICT design companies

Introduction

> It is just what is so great about this process we are in now. It is now we take all the difficult decisions. It is now we draw all these difficult lines. And then we boil and boil and boil on the exhibition concept that we can almost compare with a stew. It is not necessarily the stew that will be served. We boil and boil and boil and boil so it eventually ends up as a broth. And we include that broth as a requirement for the exhibition. So all these discussions we have been through, should we do this or that, will simply raise the level of the entire final competition. […] So it's okay that we are frustrated now and that we think it is difficult. It is part of the process. (Project manager (M), collaboration 2)

I hear the project manager use this broth metaphor several times and with good reason. He is managing the development of a new cultural historic museum and using the metaphor, he pinpoints the quintessence of collaborative design. To put together a great variety of ingredients (skills, perspectives, viewpoints, opinions) and boil them into a delicate, balanced broth. This is easier said than done. Design practice can be complex and messy (Stolterman 2008; Schön 1983) and even though this complexity and messiness might be essential for innovative and creative design to happen, it can also be overwhelming (Stolterman 2008). As the saying goes: too many cooks can indeed spoil the broth.

The aim of this paper is to explore empirically some of the contextual challenges for collaborative museum design, focusing on the interplay between museums and ICT (information and communications technology) design companies. Every day museum visitors around the world are met by audio guides, touch screens, mobile applications and so on – filled with text, sound, video, games and other content – with the intention of improving the museum experience. However, in a Danish context, many museums lack the resources and know-how needed to develop ICT. In order to go down the digital road, museums establish partnerships with ICT design companies in which curators, educators and other museum professionals collaborate with external interaction designers, creative directors and developers (Skot-Hansen 2008).

In recent years, a great deal of the literature and research concerned with collaborative design practices at museums has focused on collaboration and co-design with users and communities, for instance building on the notions of the 'participatory museum' (Simon 2010). Even though this is an important area to look into, I argue that collaborative design practices taking place between museums and ICT design companies deserve more attention. Collaborating across organisational contexts is not an easy task, but to do so and to succeed can be vital for reaching creative and innovative results (Sawyer 2007).

In what follows, I focus on two kinds of challenges for collaborative design between museums and ICT design companies: structure related challenges, and value and practice related challenges. These categories build on analysis of data from longitudinal case studies of two distinct Danish museum projects where the collaboration with ICT design companies played an essential role. Hence, the categorisation is primarily based on empirical findings and represents an initial exploration of the field.

Cases

Two distinct cases of collaboration between museums and ICT design companies serve as the basis for this study. In order to protect the anonymity of the parties involved, the cases are called 'Collaboration 1' and 'Collaboration 2' in this paper. In addition to this, the participants are named by their job title and affiliation: 'M' for museum or 'D' for ICT design company. The cases were chosen on the basis of two criteria (Neergaard 2007):

1) The cases are similar in relation to the participants' intentions regarding collaboration. In both cases, the intention was to collaborate in all phases of development of ICT, both in relation to design and construction (Löwgreen & Stolterman 2007). This contrasts some classical customer-provider relationships where, for instance, the design is developed inhouse and construction is outsourced. The criterion was applied to make sure that collaborative design actually did take place or was at least intended to take place.

2) The cases are diverse in relation to museum subject area and scale. This enables me to compare challenges in diverse settings in order to point to possible common patterns in the context.

Collaboration 1 was a collaboration between an art museum and an ICT design company. The two organisations received funding to collaborate on the development of three ICTs for three different exhibitions, the process stretching over approximately 18 months in total. The case, then, consisted of three subcases (subcase 1, 2 and 3), which provided a unique setting for the participants to reflect on and refine the collaboration on a continuous basis.

Collaboration 2 took place within a much bigger setup, namely the development of a new cultural historic museum. Hence, developing ICT was not the primary focus, but it was considered important to integrate thoughts about ICT from the outset of the development. Two phases, or subcases, in the process were particularly interesting in relation to the collaboration between museums and ICT design companies: In the first phase, an initial idea group (subcase 4) worked together over approximately 4 months with the aim of generating ideas for the museum and refining its purpose. Different creative companies and individuals, museum staff, archaeologists and other individuals with relevant knowledge participated in this phase, including an ICT design company. Later on, an exhibition design group (subcase 5) was setup to develop more concrete exhibition guidelines to be used as foundation for an architecture competition. In this process, stretching over approximately 6 months, ICT design companies were not involved directly in the group, but inputs from such companies were gathered in a more informal manner.

Method

I followed the two cases for approximately 18 months by means of ethnographic methods of observation, semi-structured interviews and document collection. My objective was generally to "follow the people" (Marcus, 1998) or more precisely the practice (Nicolini, 2009) of collaboration. I observed meetings (approx. 100 hours), produced and collected over 500 pages of field notes, documents, written correspondences and other materials and conducted 11 semi-structured interviews with selected participants.

In Collaboration 1, I conducted 7 interviews. I interviewed the project managers from the museum and from the ICT design company separately after each round, resulting in 6 interviews plus 1 interview with the overall project initiator (the project manager from the museum) after the project was complete. In Collaboration 2, I conducted 4 interviews with the project manager, the facilitator, an archaeologist from the museum and an interaction designer from an involved ICT design company after the last meeting in the idea group (subcase 4). In the interviews, the focus was on the current projects, but participants were also encouraged to compare the current projects with previous good and bad experiences of collaborations between museums and ICT design companies.

The resulting data sample was analysed with inspiration from mapping strategies where challenges are mapped and related to each other (Clarke 2006). In the mapping, two kinds of challenges stood out as central for me in relation to context. These challenges are presented and discussed next.

Challenges

Many challenges were touched upon in the data material. In this paper, I primarily draw out the ones that were specifically interesting in relation to the context of museums and ICT design companies collaborating. Also, I focus on the challenges for collaborative design to take place and not the challenges in the actual design practice of designing ICT for museums. The challenges can be categorised in two themes: Structure related challenges, and value and practice related challenges.

Structure related challenges

> Creative director (D): Both of them are mobile apps and we have been quite experimental. You find out both what works, but usually it is more important to find out what does not work. And in the last stage we will have to look at, can any of these things, any of these experiences, be reused or should we start from scratch?
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> Project manager (D): Should it be mobile?
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> Creative director (D): Should it be mobile?
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> [...]
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> Project manager (D): It could actually be good to consider this fundamentally. Because already in the first workshop [first meeting in subcase 1], we were talking about something completely different. But it had to be mobile because of the fund application. But now when it is the solution that will be implemented permanently, it might actually be good to take a step back and consider whether it really is what is wanted. (Project manager and creative director (D), collaboration 1)

In the Danish context, museums typically need funding to be able to initiate development of ICT. This seems to pose a significant structural challenge for the collaboration between museums and ICT design companies, because the museums typically have to specify some thoughts about the product and technology when applying for funding. As the project manager (M) in Collaboration 1 says:

> When you apply for funding, it is that technology infatuation that sells a project. So there are some challenges in that I think. (Project manager (M), collaboration 1)

When the actual collaboration is initiated, the application can turn out to be an unfortunate restriction. As noted in the above citation, the participants in Collaboration 1 talk about different solution formats in the very beginning of the project, but are bound to work with a mobile solution format because of the fund application. Collaborating in three cycles helps them overcome the challenge because they can argue that they have tried out the specified format in the first two rounds. In the last round they get the chance to collaborate about developing a more appropriate format.

In Collaboration 2, the structure related challenges are an even bigger issue. Much more money is involved in building a new museum and therefor the actual production has to be put out to tender. Competition rules state that competition among suppliers must be fair and all possible suppliers must be treated in the same manner. Engaging in dialogue with ICT design companies and other possible suppliers in the early design phases therefore becomes troublesome. Both the project manager and the ICT design companies involved are very attentive to the rules in Collaboration 2 and the project manager suspects the ICT design company in subcase 4 to hold back on ideas to avoid troubles with the competition rules:

> So I also think they have been holding back a little because they are afraid that if they make too much completely new thinking and put their fingerprints on it in the idea phase then they will be struck by the procurement rules. That they then cannot bid on the job in the end. They have implied that a little. (Project manager (M), collaboration 2)

In subcase 5, the project manager originally planned to involve staff from ICT design companies in the group, but partly because of the rules, he instead collects inputs from these on the side in informal chats. The project manager argues that the museum should be built from ‘the inside out’ and inputs from expertise on ICT are therefor important from the outset – both to inform an architecture competition and to be used in fund applications where, as in Collaboration 1, some thoughts about the product and technology have to be formulated. The project manager in Collaboration 2 wants to initiate a dialogue with ICT design companies and other creative industries from the outset, but this effort is challenged by the structure of procurement and competition rules.

These structure related challenges could also be relevant in other contexts of collaborative design. What makes it particularly challenging in relation to a museum context is that the mission of museum communication can be said to be to create meaningful and engaging visitor experiences (Falk &

Dierking 2013). ICT, then, has to make sense. Therefor, collaboration between museums and ICT design companies from the outset is vital to develop technology that is not just technology for the sake of technology, as an interaction designer (D) says to me:

> Technology for the sake of technology, I think that is completely wrong. You should do it because it actually gives a better experience or a better understanding of something or provides a deeper meaning that is. (Interaction designer (D), collaboration 2)

This is also crucial because museums generally have low resources for technological development and the ICT solutions therefor have to make sense and be interesting for a very long time.

Value and practice related challenges

While structure related challenges set up some barriers for initiating collaboration, value and practice related challenges surface when collaboration is actually taking place. As a project manager (D) says, two different worlds meet in the encounter:

> There is probably always the challenge. [...] that it is two worlds that need to communicate, to understand each other. And it has not been so problematic in this setup, or what should I say, as big a challenge, but there is something about understanding how each other works. (Project manager (D), collaboration 1)

Hence, we find ourselves in an arena where, in the extremes, the world of the privately owned ICT design company clashes with the world of the public museum institution, where commercialism clashes with idealism and non-profit, where high-tech and experimentation clashes with classics and conservatism, where new clashes with old. Viewed stereotypically, we are dealing with very different types of organisations and battles could be expected as these different worlds collide. However, this is far from the case in the two collaborations under study. On the contrary, there is great interest in understanding and respecting each other's worlds and learning from each other. But some challenges related to different values and practices do surface.

Firstly, in both collaborations under study, the project managers from the museums express the view that the performance of the ICT design companies does not fully live up to their expectations. In Collaboration 2, the project manager feels that the company does not bring enough original ideas to the table and he suspects that it is because of commercial reasons – they want to be able to bid on the project later on. In Collaboration 1, the project manager (M) considers the problem to arise due to lack of experience:

> It is clear that people are not very experienced in this field so therefore they promise something that they cannot actually keep. (Project manager (M), collaboration 1)

Because the technological field is ever-changing and experimental, the initial ideas or promises about technology may not always prove to be as strong as they seemed. So museums tend to have great expectations about technological creativity and possibilities, expectations that ICT design companies cannot always fulfil. But this situation may also arise because ICT design companies are very focused on the meaningfulness of technology. In both collaborations, the ICT design companies continuously stress the importance of defining the reason for technology before defining technological form. Also, they value that the good idea, the meaningfulness, is not just something they provide. They find it important that the idea is developed in collaboration. The ICT design company in Collaboration 1 see themselves as "midwifes" (creative director, collaboration 1), helping the ideas to grow and not just providing ideas. At the same time the museums seek help to think out of the box technologically, as is visible in the following citation from a meeting in Collaboration 1 (case 2):

> Exhibition developer (M): I just simply do not have the ability to think as much out of the box as we actually could, I think. So therefor it is perhaps you [looking at the staff from the ICT design company] that, to the extent that you can, have to say, 'we will do this', with consideration to some of what we want. And then we just provide the content. Because I actually think that you are much better and know a lot more about what will work and things like that.
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> Project manager (D): Well I think that would be a fine way to do it, but it's just important that you are clear about the overall purpose, so we know where we are going. (Exhibition developer (M) and project manager (D), collaboration 1)

In line with this, the ICT design companies are very conscious about focusing and simplifying when designing. In Collaboration 1, the ICT design company continuously argue that, for instance, the aim and the target group should be specifically defined, while the museum tend to want to work more openly and exploratorily with the definitions. This openness is also seen as difficult to handle in bigger

settings, especially because of the many people involved with different opinions, as an interaction designer (D) tells me:

> A general issue about working with museums is this about how many cooks are involved in a project. Which I think is a huge problem. [...] It just confuses things and decisions are made too late and people don't really know what a decision means. (Interaction designer (D), collaboration 2)

Dealing with this diversity is challenging, as the project manager in the introducing citation on page 1 points to. There is a need to boil down a lot, so that the product does not point in a thousand directions. At the same time though, the result, the broth, has to be delicate and balanced, getting the many opinions and viewpoints to mix in a tasteful manner.

This is particularly challenging in a museum context, because museum staff are generally described as idealistic and passionate about their collections and about their work. While the ICT design companies may find them very interesting to work with, they also find them full of strong, often opposing, opinions. According to the facilitator in Collaboration 2, subcase 4, museum staff are often not so used to collaboration:

> This is said with the utmost love, but museum people do not understand themselves as part of an organisation, they do not have a picture of themselves as employees with a leader for example. They understand themselves as people who go to work with passion and who have the most important jobs in the world. [...] They are not used to collaboration. (Facilitator (M), collaboration 2)

Also, museum staff are described as unused to processes where they do not have control over the outcomes, expressed here by the project manager (M) in Collaboration 1:

> But I also think it might be something to do with the expectations you have and how you understand a collaboration. And if you are afraid of being in a process and daring to let the results [pause]. Well, it is perhaps not so common in the museum world not to have control over what you are doing. (Project manager (M), collaboration 1)

So if museums are not used to collaborating and being in a process, it is indeed a challenge for the ICT design companies to collaborate and be in a process with them. This challenge may be increased by the fact that the ICT design companies museums collaborate with do not always have stable work practices. When I asked the ICT design companies about the design methods they use, it was clear that they are not used to being explicit about how they do things. As a project manager (M) tells me, she is confused about their methodology:

> They say that they work in a particular way, but then they do not do that or that is not my experience at least. The method they use is fine of course, but it is something other than what they expressed they would do. (Project manager (M), collaboration 1)

As the project manager (D) explains in the following citation, this may be because they want to treat every project as unique, but it may also be because they, as a rather new company, has not established clear working models:

> We do not have a model that we work by. For good reason because each and every project we do is different. So we do not have a toolbox we just take out and then say, 'now we do this and this and this'. It very much depends on the client and the project. And then it is perhaps also because we are a fairly new company so we have not developed a toolbox. (Project manager (D), collaboration 1)

Conclusion

To sum up, the contextual challenges for collaborative design between museums and ICT design companies could be categorised in two ways: structure related challenges, and value and practice related challenges.

Structure related challenges can set up barriers for initiating collaborative design. In the Danish context, museums typically need funding to be able to initiate development of ICT and the necessary specification about product and technology in the fund application can turn out to be an unfortunate restriction when the actual collaboration begins. Also, engaging in dialogue with ICT design companies and other possible suppliers in the early design phases can be troublesome because of competition rules when production has to be put out to tender. This is particularly challenging in a museum context because the use of technology has to be meaningful to match the general mission of museum communication. To define this meaningfulness knowledge about both the museum and ICT is needed.

While structure related challenges can set up barriers for initiating collaboration, value and practice related challenges surface when collaboration is actually taking place. On one hand, museums have

great expectations regarding technological creativity and possibilities, and they seek help from ICT design companies to think out of the box technologically. On the other hand, ICT design companies stress the importance of defining the reason for technology before defining technological form. Furthermore museum staff are described as not being used to collaboration and to being in a process, while the ICT design companies they collaborate with are not explicit about their methodology. While museums might need some directions about designing collaboratively, ICT design companies might not be so good at explicating such directions.

In regards to both kinds of challenges, it is crucial how interaction is managed and dialogue facilitated across the organisational contexts. To return to the citation in the very beginning of this paper, it is the quintessence of collaborative design to get the right ingredients and to make sure they are boiled together tastefully so the broth will not end up being spoiled. In this paper, the focus has been on exploring the challenges for collaborative design and I have only touched lightly upon how the challenges could be met. Further work needs to be done to examine this more thoroughly. Also, it could be interesting in further research to compare other cases in both a Danish context and more international contexts.

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